

CS/CE 355/373 DATABASE SYSTEMS

Course Project

You are required to develop a course project focused on automating data processes. Your objective is to build an information system capable of efficient data storage and retrieval. Importantly, this project must cater to various user roles, encompassing administrators, power users, and general users. Each user role will have a different level of capability within the system. A list of potential projects is given at the end.

The project should incorporate a minimum of five essential functional requirements along with at least one transaction processing screen—an interface allows the implementation of one complex business process such as completing a sales transaction for an online store. Furthermore, we expect your project would have at least three setup screens to ensure the effective configuration of the system.

The project has a 15% weightage in the total grade and will be evaluated in four milestones. The rubric for each milestone will be released separately.

Milestone	Weightage
Proposal Submission	10%
Interim ERD	20%
Interim Demo	30%
Final Presentation and Submission	40%
Total	100%

Proposal Submission:

You are provided a template for the Project Proposal. Each group (2-3 students) has to submit one proposal with all the required information. You are provided with sample good, average, and worst proposals for your reference. Discussing your project proposal with your instructor or course staff before submission is recommended.

The rubric for Proposal Submission is given below:

Category	Description	Weightage
Project Description and Application	How effectively communicates project goals and scope.	2%
	How is the project useful in addressing real-world issues and what impact can it potentially have?	
Functional Requirements	Clear, comprehensive feature list aligns with objectives, accuracy in catering to user needs.	4%
Screens	Clear UI design, layout coherence, effective user interaction showed	4%

Total		10%
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Sample Projects:

Note: You may not need to implement an entire system; instead, you could choose to implement specific modules. Discussing with your instructor or course staff could help you identify the necessary modules to implement for your project.

- **Hospital Management System:** A software to streamline healthcare operations, manage patient records, appointments, billing, and medical history for efficient hospital administration.
- **Inventory Management System:** A tool to track and manage inventory levels, orders, and stock movement, ensuring optimal stock control and minimizing shortages.
- **Criminal Database System:** A database to store and manage criminal records, aiding law enforcement agencies in tracking criminal history and solving cases.
- **Student Club Management System:** A platform to manage student clubs and their activities, facilitating communication, event organization, and membership tracking.
- **Manufacturing Management System:** Software designed to oversee manufacturing processes, production planning, resource allocation, and quality control in industrial settings.
- **Accounting System:** A software for recording and managing financial transactions, generating reports, and maintaining accurate financial records for businesses.
- **Supply Management:** A system to manage the flow of goods and services, from procurement to distribution, optimizing supply chain efficiency and reducing costs.
- **Time Table Management System:** A tool to create and manage schedules for classes, meetings, or events, ensuring effective time utilization and avoiding conflicts.
- **Human Resource Management System:** Software to manage employee information, payroll, attendance, and recruitment, simplifying HR tasks and improving workforce management.
- **Tournament Management System:** A platform to organize and manage tournaments, from registrations and scheduling to score tracking and results dissemination.
- **Customer Relationship Management System:** Software to track and manage interactions with customers, enhancing customer service, sales, and marketing efforts.
- **Issues/Ticket Management System:** A system to receive, track, and resolve customer complaints, technical issues, or service requests, ensuring efficient problem resolution.